Ji Xie

jixie@andrew.cmu.edu • horizonwind2004.github.io • Google Scholar

EDUCATION

Carnegie Mellon University, Language Technologies Institute *08/2026-Present*
PhD Student; Advisor: Louis-Philippe Morency; Expected graduation: May 2031

UC Berkeley, Berkeley AI Research (BAIR) *03/2025-12/2025*
Visiting Student; Hosts: Trevor Darrell, XuDong Wang

Zhejiang University, Chu Kochen Honors College *09/2022-06/2026*
BEng in Computer Science and Technology; **GPA: 4.20/4.3 | Rank: 2/148**

RESEARCH INTEREST

My research focuses on **scalable, unified, controllable multimodal models**, with a long-term goal of developing **world models for embodied intelligence**.

INDUSTRY EXPERIENCE

Research Intern, ByteDance Seed *10/2025-08/2026*

- **Seedream 5.0 series (Lite / Pro):** Core contributor to interactive precision editing.
- **Led precise color control and palette-guided image generation and editing**, supporting Hex color codes and external color swatches.
- Core contributor to **layer separation, text edit, and precise local editing**.

Mentor: Xun Wang.

SELECTED PUBLICATIONS

- **Ji Xie**, Trevor Darrell, Luke Zettlemoyer, XuDong Wang. "Reconstruction Alignment Improves Unified Multimodal Models." *ICLR 2026*. Code.
- Jiaxin Ge*, Yiming Qin*, **Ji Xie**, Haozhe Jiang, Xiaochuang Han, Junyi Zhang, Andrew Dai, Yinfei Yang, Jitendra Malik, Ranjay Krishna, Sewon Min, Haiwen Feng, Le Xue, Baifeng Shi, Trevor Darrell, XuDong Wang. "OmniTaskonomy: When Does Visual Generation Improve Visual Understanding?" *arXiv, 2026*.
- **Ji Xie**, Yanhao Jia, Xiangpeng Yang, Martin Q. Ma, Sihan Xu, Erik Cambria, Louis-Philippe Morency. "Improving and Expanding Video Editing without Any Video." *In submission*.
- **Ji Xie**, Dewei Zhou, Xinyu Huang, Zhennan Chen, Xun Wang. "Paint-Anything: Unified Any-Color Control for Image Generation and Editing." *NeurIPS 2026 / Seed Tech Report.*
- Xiangpeng Yang, **Ji Xie**, Yiyuan Yang, Yan Huang, Min Xu, Qiang Wu. "VideoCoF: Unified Video Editing with Temporal Reasoner." *CVPR 2026; Highlight*. Code.
- Zechuan Zhang, **Ji Xie**, Yu Lu, Zongxin Yang, Yi Yang. "In-Context Edit: Enabling Instructional Image Editing with In-Context Generation in Large Scale Diffusion Transformer." *NeurIPS 2025*. Code.
- Dewei Zhou*, **Ji Xie***, Zongxin Yang, Yi Yang. "3DIS: Depth-Driven Decoupled Instance Synthesis for Text-to-Image Generation." *ICLR 2025; Spotlight*. Code.

* Equal contribution.

INVITED TALKS

Apple Research *10/2025*
"Reconstruction Alignment Improves Unified Multimodal Models."
Hosted by Chen Chen and Yinfei Yang.

SELECTED COMPETITION & AWARDS

SenseTime Scholarship (**top 30 recipients annually in China**) *06/2025*
International Collegiate Programming Contest (ICPC), Shenyang Site, 2022, **Gold Medal** *10/2022*
China Collegiate Programming Contest (CCPC), Guangzhou Site, 2022, **Gold Medal** *10/2022*

PROFESSIONAL SKILLS

Core Stack: Python, C/C++, PyTorch, NumPy, Matplotlib, Linux. **Codeforces:** 2478 (Epyset).